

1.2. nešpre $\Leftarrow$

line tedj je $a_{11} > 0$ a $\det(A) > 0$ nebol $a_{11} \cdot a_{22} - a_{12}^2 > 0$

overine je $\langle, \rangle$ skalarne definicije s.s.

$$\langle u, v \rangle = v^T A v = (u^T A v)^T = v^T A^T u = v^T A u = \langle v, u \rangle = \overline{\langle v, u \rangle}$$

Linearita

$$\langle u, t \cdot v \rangle = u^T A (t \cdot v) = t \cdot v^T A v = t \langle u, v \rangle$$

$$\langle u, v+w \rangle = u^T A (v+w) = v^T A v + u^T A w = \langle u, v \rangle + \langle u, w \rangle$$

Definitna pozitivna

$$\text{Bod } u = \begin{pmatrix} x \\ y \end{pmatrix} \quad A = \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix}$$

$$\begin{aligned} \langle u, u \rangle &= u^T A u = \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = a_{11} x^2 + 2a_{12} xy + a_{22} y^2 \\ &= \frac{(a_{11}x + a_{12}y)^2}{a_{11}} + y^2 (a_{11}a_{22} - a_{12}^2) \rightarrow \det(A) \text{ (kladi)} \end{aligned}$$

$\rightarrow$ (kladi)

teno ugla je zveš kladj a nalogj pozro pokid

$$y = 0$$

$$a (a_{11}x + a_{12}y) = 0$$

$$\text{Tedy } \begin{pmatrix} x \\ y \end{pmatrix} = \vec{0}$$

$\Rightarrow \langle, \rangle$ spliča DP a tedj $\langle u, u \rangle \geq 0$ a $\langle u, u \rangle = 0$ a $\langle u, u \rangle = 0$

$$\langle u, u \rangle = \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = a_{11} x^2 + 2a_{12} xy + a_{22} y^2$$

Pokid $a_{11} = 0$ tak $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ by nelo $\langle u, u \rangle = 0$ SPOT.

tedy $a_{11} \neq 0$

tudi:

$$\langle u, u \rangle = \frac{(a_{11}x + a_{12}y)^2 + y^2(a_{11}a_{22} - a_{12}^2)}{a_{11}}$$

Kdyby $a_{11} < 0$ pak $\left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\rangle < 0$

Kdyby $(a_{11}a_{22} - a_{12}^2) = \det A \leq 0$ tedy $a_{11} > 0$

$$u = \begin{pmatrix} -a_{12} \\ +a_{11} \end{pmatrix}$$

SP/kuže $\langle u, u \rangle = a_{11} \det(A) \leq 0$

ale $\begin{pmatrix} -a_{12} \\ a_{11} \end{pmatrix}$ není $\vec{0}$.

tudi:

$\det A > 0$